Economics 5040: Dynamic Stochastic Modeling
Midterm Examination — Answer Key

Fall 2026

1. Inter-temporal Prices

A higher real interest rate raises the amount of future consumption that can be obtained by giving up one unit of consumption today, so current consumption becomes more expensive relative to future consumption. The substitution effect therefore gives the household an incentive to consume less today, save more, and shift consumption toward the future.

2. Choice Under Uncertainty

Expected utility from the gamble is

$$\frac{1}{2}\sqrt{0} + \frac{1}{2}\sqrt{100} = 5.$$

The certainty equivalent CE solves $\sqrt{CE} = 5$, so $\boxed{CE = 25}$; thus the gamble is worth \$25 to her.

3. Markov Processes

The element P_{ij} is the conditional probability that next period's state is j, given that the current state is i. Row i therefore gives the complete probability distribution over next period's states conditional on being in state i today.

4. Introduction to Dynamic Programming

The Bellman equation says that, given current state x, the decision maker chooses u to maximize the sum of the current reward $r(x, u)$ and the discounted expected value of the future. The continuation value $\beta E[V(x') \mid x, u]$ captures how today's action affects the distribution of tomorrow's state and therefore all future opportunities.

5. Introduction to Dynamic Programming

The value function $V(x)$ gives the maximum attainable discounted value starting from each state x. The policy function gives the optimal action u to take in each state, so the value function tells us how well the decision maker can do while the policy function tells us what the decision maker should do.

6. Howard's Method versus Value Function Iteration

Value function iteration makes only one Bellman update before reconsidering the policy, so information about the consequences of a candidate policy propagates gradually through the value function. Howard's method evaluates a fixed policy much more fully before improving it, allowing the algorithm to make larger, better-informed policy improvements and usually reducing the number of iterations dramatically.

7. The Solow Model

A permanently higher saving rate raises investment, so the economy converges to a higher steady-state level of capital and therefore a higher level of output. It creates transitional growth while the economy moves to the new steady state, but it does not permanently raise the long-run growth rate of output per worker in the basic Solow model.

8. Optimal Saving

The Euler equation says that at an interior optimum the marginal utility lost by giving up one unit of consumption today equals the expected discounted marginal utility obtained from the R units of consumption that saving makes available tomorrow. If the right-hand side were larger, saving would have the greater marginal value, so the household would reduce current consumption and save more.

9. Income Fluctuations

With concave utility, low-consumption outcomes have especially high marginal utility, so uncertainty about future income makes assets valuable as insurance against those bad states. The household may therefore engage in precautionary saving, accumulating more assets today even though expected future income is unchanged.

10. Putting the Ideas Together

A job search problem is dynamic because today's accept-or-reject decision determines whether the worker remains unemployed and searches again or enters employment, thereby changing tomorrow's state and future opportunities. A natural state includes current employment status and the current wage offer (together with any other payoff-relevant information), the choice is whether to accept or reject the offer, and rejecting typically returns the worker to the unemployed/search state while accepting moves the worker into employment.

Suggested Grading

Each question can be graded on a 0–5 scale. A score of 5 identifies the central concept and explains the relevant economic mechanism clearly; 4 is essentially correct with a minor omission; 3 shows the basic idea but gives an incomplete or imprecise explanation; 2 shows partial understanding with an important error or omission; 1 shows minimal relevant understanding; and 0 is incorrect or unanswered.